
Chapter 4: Quadrilaterals

Worksheet 4: Properties of Quadrilaterals

Learning Objective: Students will identify and classify quadrilaterals (parallelogram, rhombus, rectangle, square, trapezium, kite); apply their properties to find unknown angles and sides; prove properties using angle-sum relationships; and solve real-life problems.

Worked Example

The angles of a quadrilateral are in ratio $2 : 3 : 5 : 8$. Find each angle.

Solution: Sum of angles in a quadrilateral = 360° .

$$2k + 3k + 5k + 8k = 360^\circ \Rightarrow 18k = 360^\circ \Rightarrow k = 20^\circ$$

Angles: $40^\circ, 60^\circ, 100^\circ, 160^\circ$.

Questions 1–5: Easy / Recall

Q1. What is the sum of interior angles of any quadrilateral?

Q2. State two properties of a parallelogram.

Q3. A rectangle has one angle equal to 90° . What are all four angles?

Q4. In a rhombus, one side is 8 cm. What are the other three sides?

Q5. Name a quadrilateral with exactly one pair of parallel sides.

Questions 6–10: Basic Application

Q6. Three angles of a quadrilateral are 75° , 90° , 110° . Find the fourth.

Q7. In a parallelogram, one angle is 65° . Find all four angles.

Q8. The diagonals of a rhombus are 16 cm and 12 cm. Find the side of the rhombus.

Q9. In a rectangle, the diagonal is 17 cm and one side is 8 cm. Find the other side and perimeter.

Q10. A kite has two pairs of equal adjacent sides: 5 cm and 8 cm. Find its perimeter.

Questions 11–15: Practice and Skill Building

Q11. Prove that opposite angles of a parallelogram are equal. (Use angle-sum property.)

Q12. In a parallelogram $ABCD$, $\angle A = 3x + 10^\circ$ and $\angle C = 5x - 30^\circ$. Find x and all angles.

Q13. The perimeter of a rhombus is 52 cm. Its one diagonal is 24 cm. Find the other diagonal and its area.

Q14. In a trapezium $ABCD$, $AB \parallel CD$. $\angle A = 70^\circ$ and $\angle B = 110^\circ$. Find $\angle C$ and $\angle D$.

Q15. Which quadrilaterals have diagonals that bisect each other? Which have diagonals equal in length?

Questions 16–18: Higher Order Thinking

Q16. $ABCD$ is a parallelogram. P is the midpoint of AB and Q is the midpoint of CD . Show that $APCQ$ is also a parallelogram.

Q17. A square and a rhombus have the same perimeter of 40 cm. Compare their areas if the rhombus has diagonals in ratio 3:4.

Q18. The sum of two adjacent angles of a parallelogram is 180° . Use this to show that if one angle is a right angle, all angles must be right angles, making it a rectangle.

Questions 19–20: Real-Life Applications

Q19. A farmer's field is shaped like a parallelogram with base 120 m and height 80 m. Find the area. Fertiliser costs Rs. 15 per sq. m. Find the total cost.

- Q20.** The floor of a room is shaped like a trapezium with parallel sides 14 m and 10 m and height 8 m. Find the floor area. Tiles of $0.5\text{ m} \times 0.5\text{ m}$ cost Rs. 40 each. Find the cost of tiling the floor.

Reflection Question: A square is a special rhombus and also a special rectangle. Explain how. Draw a Venn diagram of quadrilateral types. (Teacher Verification Required)